

Quantum-Enhanced Deep Q Learning with Parametrized Quantum Circuit

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Abstract—Quantum Computing (QC) is fastly growing to replace current computing systems for many high-performance applications. Due to the rapid increase of computational power, machine learning models based on artificial neural networks (ANN) have become highly effective. Even though classical machine learning models have been performing well, quantum computing with machine learning, i.e., Quantum Machine Learning (QML), will enhance the performance in multiple ways. Subsequently, Deep Q learning is a prominent approach to reinforcement learning used in solving complex applications for a desired performance. But this performance can be improved using a quantum computer. This paper proposes a quantum-enhanced Deep Q Learning algorithm to see its advantages over its classical counterparts. We implement our approach using parametrized quantum circuit (PQC). The evaluation of the proposed method achieves the metrics rewards and episodes up to 500 points and 1200, respectively. The performance shows that it collects roughly four times more rewards in a given time and takes significantly fewer episodes to converge compared to the classical approach.

Keywords—Quantum Computing; Quantum Machine Learning; Quantum Reinforcement Learning; Deep Q Learning; Parametrized Quantum Circuit.

I. INTRODUCTION

With the advancement of recent VLSI technology, a slowdown in Moore's Law is seen with the latest computing systems. An observation made by Moore in 1965 states that the number of transistors per unit area of an Integrated Circuit (IC) will double roughly once every two years. Moore's law has existed since it was stated, leading to our fantastic, powerful computers. But it is reaching its saturation point. Most experts believe it will saturate entirely by 2025 [1].

Transistors get smaller than their physical limit with the continuous reduction in the feature size. Alternatively, quantum mechanics satisfactorily describe the behavior of electrons. This hints to look into quantum computers and quantum information processing. However, classical computers are still the best for jobs currently handled. With the saturation of Moore's Law, the computing power of these computers will tend to saturate. Quantum computers will be superior in this place, explicitly solving complex problems such as algorithmic tasks using quantum mechanical phenomena-superposition and entanglement, and many classical intractable problems [2]. Deep reinforcement learning, which takes advantage of the power of artificial neural networks, has opened new successes,

such as going beyond human limits in challenging games, e.g., Go and Chess. The domain has already become an unstoppable flow of artificial intelligence (AI) research.

Cutting-edge quantum computing technology advances to realize the quantum computational benefit over modern computers while anticipating numerous algorithms [3]. Additionally, it is now possible to include these quantum calculations into existing machine learning algorithms by developing a variational quantum circuit (VQC), miming the concepts and operations of artificial neural networks. This has emerged as a significant trend in quantum machine learning, and numerous experiments, including variational classifiers, image preprocessing, federated learning, etc., are carried out using a VQC. In this case, the circuit acts as a quantum neural network (QNN) [4]. The field of quantum machine learning enhances machine learning via quantum computing techniques. It has garnered much attention, especially in the reinforcement learning research community. The quantum-enhanced paradigm known as Quantum Reinforcement Learning (QRL) has a lot of promise. Subsequently, recent works in the area include [4]–[6].

Quantum computers have the potential to revolutionize computations on specific complex applications. A quantum computer with machine learning may provide potential impacts on these real-world applications, i.e., Quantum Machine Learning (QML) will enhance the performance in multiple ways [1]. For example, Deep Q Learning (DQL) is a prominent approach to reinforcement learning used in solving complex applications for a desired performance. This performance can be further improved using a quantum computer. This paper proposes a variant of the DQL algorithm given to a quantum computer and observes its performance over a classical approach. The proposed quantum-enhanced DQL is implemented with a parametrized quantum circuit. Experimental results show that the proposed system achieves the metrics rewards and episodes up to 500 points and 1200, respectively. This performance is 4× than achieved with a classical approach in a given time. Further, the proposed method takes 400-600 episodes less to converge than its classical counterpart.

The rest of the paper is organized as follows. Section II states literature. Section III briefs about quantum-enhanced reinforcement learning. Section IV describes the proposed approach. Section V talks about the results obtained after running the proposed model. Section VI concludes the paper.

II. RELATED WORKS

The field of quantum machine learning enhances machine learning via quantum computing techniques. It has garnered much attention, especially in the reinforcement learning research community. The quantum-enhanced paradigm known as Quantum Reinforcement Learning (QRL) has a lot of promise. Subsequently, recent works in the area include [4]–[9]. Greff *et al.* [4] discussed the use cases of several variants of the long-short-term memory (LSTM) architectures. They also applied these architectures to practical problems like handwriting recognition and music modeling. Through repeated interactions with its environment, an agent or a group of agents maximizes its long-term return using the reinforcement learning (RL) machine learning technique [7]. Ye *et al.* [8] presented a deep reinforcement learning-based approach for resource allocation in a vehicular communication network. The authors showed the mapping between the local observations, such as local channel state information (CSI) and interference levels, and the resource allocation and scheduling solution. Jerbi *et al.* [9] represented near-term quantum computers as hybrid quantum machine learning models to establish a hypothesis family for a specific learning task using parametrized and data-dependent quantum computations via classical optimization. Parametrized quantum circuits (PQCs) have already demonstrated their effectiveness in classification, generative modeling, and clustering issues. Such circuits were employed to train these learning models. Skolik *et al.* [5] explained the complex interplay between the environment's specification and the demands placed on the quantum model's readout operators. They demonstrated how a quantum DQL agent can only be successful if these conditions are satisfied. Chen *et al.* [6] offered a strategy for RL agents to optimize the quantum circuit parameters, which are evolutionary and gradient-free. They demonstrated how their quantum deep RL model attains a state-of-the-art result on the Cart-Pole problem with just a few parameters. Figure 4 describes the variational quantum circuit for reinforcement learning. Here $|0\rangle$ represents the input qubit, R stands for a rotation gate, and Q stands for a superposition gate. Schneck *et al.* [10] suggested a hybrid actor-critic RL algorithm at cern beamlines. Reinforcement learning (RL) and deep reinforcement learning (DRL) are two paradigms of complex sequential decision-making systems and satisfy the requirements of automatic policy learning under uncertainty. Yang *et al.* [11] focused on the empowerment of DRL on noisy intermediate state quantum (NISQ) computation, which refers to an agent interacting with the environment to gain knowledge of backgrounds and derive the policy of decision-making accordingly. Wei *et al.* [12] discussed how intelligent agents accelerate overall quadratic speedup by increasing the training performance of DRL organically without intentional hyperparameter adjustment. The authors presented a quantum-inspired experience replay (QER) approach for DRL (DRL-QER) to gain speedup.

III. QUANTUM ENHANCED REINFORCEMENT LEARNING

Reinforcement learning (RL) is one of the fundamental paradigms of machine learning, along with supervised and

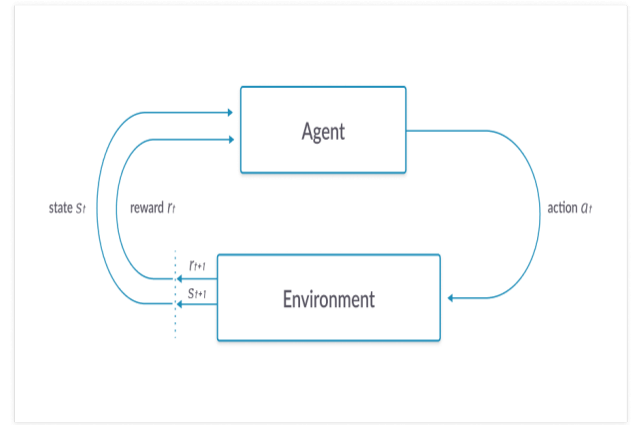


Fig. 1: Approach of Reinforcement Learning

unsupervised learning. In this paradigm, an intelligent agent learns how to take action in an environment to maximize the reward it receives. It consists of both exploration of unknown territory and exploitation of present information. An RL agent performs the action in an environment for which it receives a reward/punishment. Based on the reward/punishment, the agent learns to make future decisions. Reinforcement learning has been successfully applied to many real-world applications like computer vision, self-driving, etc. Although classical reinforcement learning has succeeded, researchers are now looking to enhance reinforcement learning algorithms by applying quantum computing methods [13]. Figure 1 describes the general setting of reinforcement learning.

Quantum reinforcement learning is a promising paradigm that could accelerate machine learning calculations. It aims to design and build quantum agents to exchange information with their environment and adapt to achieve a goal. The goal is often to maximize the reward. Learning is often accelerated with a quantum algorithm. For example, Grover's search algorithm is usually applied to obtain a quadratic speedup in the learning process.

IV. PROPOSED WORK

A. Deep Q Learning

Deep Q-learning is a kind of reinforcement learning that approximates the Q-function, which calculates the expected future benefits of performing a specific action in one particular state using a neural network. The algorithm operates by repeatedly performing activities in a given environment, monitoring the ensuing state and reward, and then modifying the Q-function following the discrepancy between the expected and real rewards. The Bellman equation which iteratively determines the ideal Q-value for a specific state-action combination, serves as the foundation for the update rule, as shown in Algorithm 1. The agent uses an epsilon-greedy policy to choose an action at each time step that strikes a balance between exploitation and exploration. Atari games and robotic navigation tasks are only two examples of the complicated control issues that the algorithm has been proven to be successful in resolving.

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Initialize replay memory  $D$ ; Initialize Q-network with
random weights  $\theta$ ; for  $episode = 1$  to  $M$  do
  Initialize state  $s_1$ ; for  $t = 1$  to  $T$  do
    With probability  $\epsilon$  select a random action  $a_t$ ,
    otherwise select  $a_t = \text{argmax}_a Q(s_t, a; \theta)$ ;
    Execute action  $a_t$  in the environment and
    observe reward  $r_t$  and next state  $s_{t+1}$ ; Store
    transition  $(s_t, a_t, r_t, s_{t+1})$  in  $D$ ; Sample a
    mini-batch of transitions  $(s_j, a_j, r_j, s_{j+1})$ 
    from  $D$ ; Calculate the target Q-value for each
    transition; Update Q-network parameters by
    minimizing the loss:
     $\mathcal{L}(\theta) = \frac{1}{N} \sum_{j=1}^N (y_j - Q(s_j, a_j; \theta))^2$ ; Every
     $C$  steps, copy Q-network weights to target
    network:  $\theta^- \leftarrow \theta$ ;
  end
end

```

Algorithm 1: Deep Q Learning

B. Proposed Quantum-Enhanced Deep Q Learning

Deep Q learning is one of the approaches to reinforcement learning where the agent tries to make decisions in the environment by updating its approximation of the Q-Function. Q-Function is the model of the environment the agent has built by interacting with the environment so far. We propose using Parametrized Quantum Circuits as Q-function approximators for the Deep Q Learning algorithm. Figure 2 provides an overview of the Variational Quantum Circuit for reinforcement learning.

At the core of this Reinforcement Learning algorithm is a PQC that takes the agent's states in the environment (i.e., a numpy array) as input and outputs a vector of expectation values. These expectation values are then post-processed to approximate Q-values $Q(s, a)$. Q values compare how promising the future rewards will be if the agent takes action 'a' at state 's'. In this way, the PQCs play an analog role to deep neural networks in modern RL algorithms. A popular way to encode an input vector in a PQC is through single-qubit rotations, where the components of this input vector control rotation angles. To get a highly-expressive model, these single-qubit encodings are not performed only once in the PQC but in several "re-uploadings", interlayered with variational gates. Figure 3 shows a simple circuit on three qubits obtained by composing various single qubit rotations. The variational quantum circuit is shown in Figure 4. Here $|0\rangle$ represents the input qubit, R stands for a rotation gate, and Q stands for a superposition gate.

Quantum Enhanced Deep Q-learning method uses a PQC (shown in Figure 3) to approximate the Q-function of the agent defined in Equation 1. Here $\langle O_a \rangle_{s, \theta}$ are expectation values of observables O_a (one per action) measured at the output of the PQC.

$$Q_{\theta}(s, a) = \langle O_a \rangle_{s, \theta} \quad (1)$$

Algorithm 2 describes the Q function approximator using a PQC. The algorithm works as follows; it trains a parametrized quantum circuit (PQC) for a reinforcement learning task. The

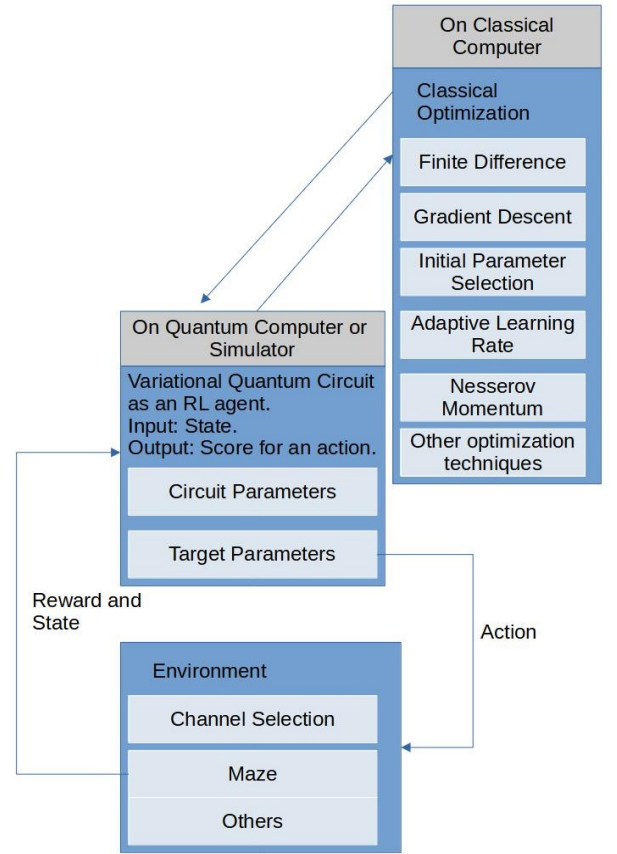


Fig. 2: Overview of Variational Quantum Circuit for Reinforcement Learning

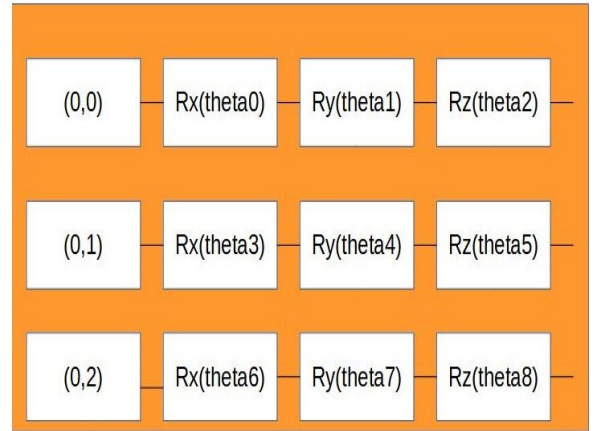


Fig. 3: PQC Circuit generated by composing various single qubit rotation operations.

algorithm uses a replay memory to store past experiences and samples a batch of 1-step interactions to update the PQC parameters via gradient descent. The PQC is used to estimate the Q-value function, which represents the expected future reward of taking an action in a given state. The algorithm uses an ϵ -greedy policy to select activities based on the estimated Q-values. The target Q-values are calculated using a separate PQC with target parameters periodically updated from the main PQC parameters. The loss function is the mean squared error between the predicted and target Q-values, which is minimized during the parameter update. The algorithm terminates after a set number of episodes. After

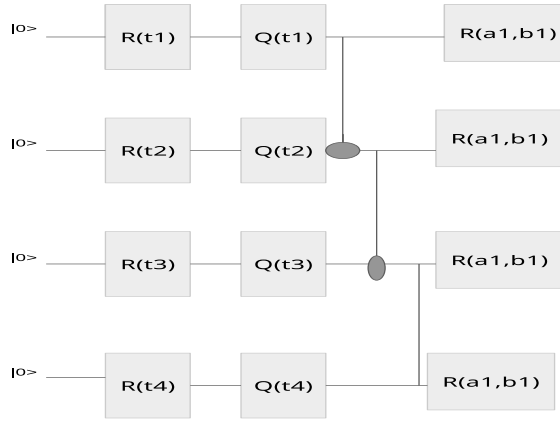


Fig. 4: Proposed variational Quantum Circuit for Reinforcement Learning

Data: replay memory $\mathcal{M}$, discount factor γ , PQC parameters θ , target PQC parameters θ'
Result: trained PQC parameters θ
initialize replay memory $\mathcal{M}$;
initialize PQC with parameters θ ;
initialize target PQC with parameters θ' ;
for $episode = 1$ **to** N **do**
 initialize environment and observe initial state S ;
 while $episode$ is not terminated **do**
 select action a using an ϵ -greedy policy based on $Q_\theta(s, a)$;
 execute action a and observe reward r and next state S' ;
 add (S, a, r, S') to replay memory $\mathcal{M}$;
 sample a batch $\mathcal{B}$ of 1-step interactions (S, a, r, S') from $\mathcal{M}$;
 calculate target Q-values for batch using target PQC parameters θ' :
 $y_j = r_j + \gamma \max_{a'} Q_{\theta'}(S'_j, a')$;
 calculate predicted Q-values for batch using PQC parameters θ : $Q_j = Q_\theta(S_j, a_j)$;
 update PQC parameters θ using gradient descent to minimize the loss function:
 $\mathcal{L}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{j \in \mathcal{B}} (y_j - Q_j)^2$;
 sporadically update target PQC parameters θ' from main PQC parameters θ ;
 set $S = S'$;
 end
end

Algorithm 2: Deep Q-Learning with PQC

developing this approximator, one can implement a Deep Q Learning algorithm. An exploration policy which how much weight an agent gives to recent interactions compared to the older interaction. So, the policy is chosen cleverly. For the exploration policy of the agent, we use ϵ -greedy policy as defined in Equation 2. Note that the ϵ value is multiplicatively decayed at each interaction episode.

$$\pi(a | s) = \begin{cases} \delta_{a, \arg\max_{a'} Q_\theta(s, a')}, & p=1-\epsilon \\ \frac{1}{\text{num_actions}}, & p=\epsilon \end{cases} \quad (2)$$

C. Estimation of the State-Value Function

Monte Carlo methods learn from complete sample returns, which implies that they are defined for episodic tasks only. An update happens after each episode, so learning happens directly from experience. The goal is to learn the optimal policy given some episodes, and the basic idea is to average the returns after a state s is visited. A distinction is made about whether the returns for each visit to s in an episode are averaged or only first-time visits.

The first illustration uses traditional Monte Carlo policy assessment to produce n sub-optimal state-value functions. We want to emphasize that the formulation of the algorithm allows for either the embedding of n classically generated (sub-optimal) state-value processes for approximating an improved state-value function or the generation of quantum-enhanced state-value functions by feeding the algorithm with episodes' individual (non-averaged) rewards. A nested algorithm may also incorporate both of these. The preferred state-action combinations will be determined by the QUBO formulation based on their rewards across the episodes, and when this has been done n times for m episodes, n state-value functions can be produced using the same procedure. Based on m episodes, identify n sub-optimal state-value functions classically, embed each state-action pair with its corresponding value directly on the QPU, and thus contribute to approximating the optimal state-value function, which is superior to each of the n sub-optimal state-value functions.

V. EXPERIMENTAL RESULTS

To check the effectiveness of the proposed approach, we implement it by choosing two environments from the OpenAI GYM library. These environments, Cartpole and Acrobot, belong to the classic control genre. We measure the performance of the proposed model concerning these environments and compare the performance metrics with the classical approach.

A. Experimental Setup

The experiments are conducted using the popular machine learning library tensor flow and the TensorFlow quantum machine learning version. The proposed Deep Q model is built using tensorflow keras. The parameterized quantum circuits are built using the "cirq" library. The optimizer used during the experiment is Adam from the Keras package. The experiments are run on a google hosted runtime service provider with Tensor Processing Unit (TPU) accelerated hardware. Standard Cartpole and Acrobot environments from openAI GYM are used to evaluate the model's performance.

B. CartPole-Version1

1) *Environment Description:* An unactuated joint attaches a pole to a cart, oscillating along a smooth track. The pendulum is placed upright on the cart, and the goal is to balance the pole by applying forces to the left and right sides of the cart. The observation space has four variables: position, speed, angle, and angular speed. The goal is to keep the pole upright. A reward of +1 is added for each action until the pole is upright. The action space has two actions that apply a force from left or right.

2) *Model Description*: The description of the environment plays a significant role in building the model. The dimension of the observation space determines the number of inputs to the PQC. Similarly, the size of the action space (number of possible actions the agent can perform) determines the number of outputs. A PQC with four inputs (Dimension of the Observation Space) and two outputs (Dimension of the Action Space) are created to implement the proposed quantum-enhanced Deep Q Learning approach. Figure 5 describes a sample configuration of the model parameters.

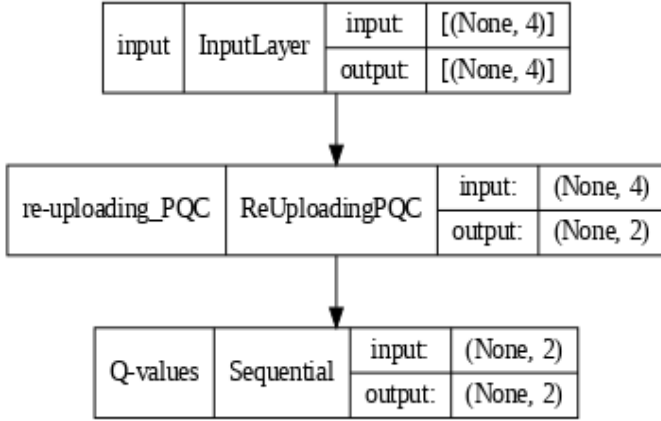


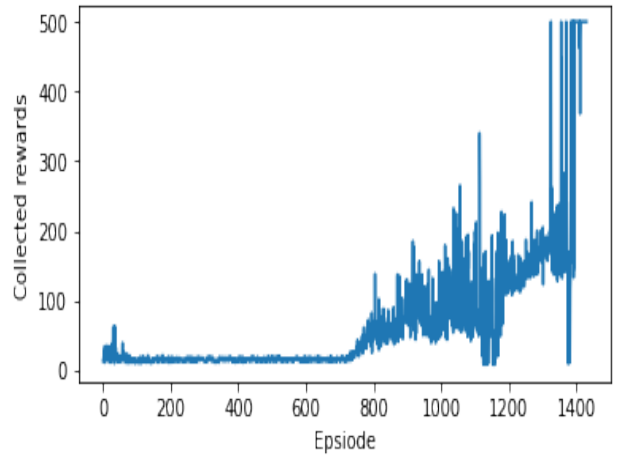
Fig. 5: PQC for Cartpole Q-function approximation.

3) *Model Performance*: The metric we use to measure the model's performance is the average rewards gained per 10 episodes. This is a suitable metric because it balances out the episodes which are outliers. The plot shows the performance of quantum deep Q-learning on the CartPole environment, a popular benchmark in reinforcement learning. The y-axis shows the episode reward, the cumulative reward obtained by the agent during a single task episode, while the x-axis shows the episode number. The algorithm achieves an average reward of approximately 200 after around 500 episodes, which is the maximum achievable reward in this environment. The plot also shows some variability in the reward obtained by the agent across different episodes, which is expected due to the environment's stochastic nature and the algorithm's exploration strategy. Overall, the plot suggests that the quantum deep Q-learning algorithm can learn to solve the CartPole task effectively.

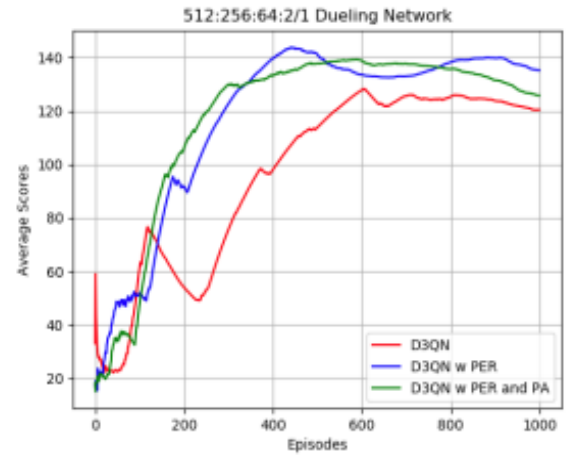
4) *Comparison Study*: The proposed model, even with a minimal circuit compared to that of a complex Deep Q network of the classical approach, performs better than the classical Deep Q Learning as evident by Figure 6b. Even though the initial performance of the classical approach is better until 800 episodes, the proposed method reaches a much higher reward of 100 to 500 for the episodes in the range of 800-1400. In this episode range, the classical approach starts to saturate and eventually declines after reaching a reward of 140.

C. Acrobot

1) *Environment Description*: The system consists of two links connected to form a chain, with one end of the chain



(a) Average rewards by proposed DQL approach on Cartpole environment



(b) Deep Q Learning for Cartpole environment

Fig. 6: Average score by the proposed and classical approach on cartpole environment

fixed. There is an actuated joint between the two links. The goal is to apply torques on the actuated joint to swing the free end of the straight chain above a fixed height beginning from the starting state of facing downwards. The dimension of observation and action space are six and three, respectively. The goal is to have the free end reach a designated target height in as few steps as possible, and as such, all actions that do not get the goal incur a reward of -1. Achieving the target height leads to termination with a reward of 0.

2) *Model Description*: A PQC with six inputs (Dimension of the Observation Space) and three outputs (Dimension of the Action Space) are created to implement the quantum-enhanced Deep Q Learning approach. Figure 7 describes the model parameters.

3) *Model Performance*: Along with average rewards gained per 10 episodes, we additionally measure the number of steps(interactions) the agent takes to reach the goal. Figure 8a indicates the average rewards collected per episode. The last and best contained are -500 and -380, respectively. This shows the agent achieved a maximum score of 120. As shown in Figure 8b, the number of steps the agent requires to reach the goal gradually decreases. If the agent does not match

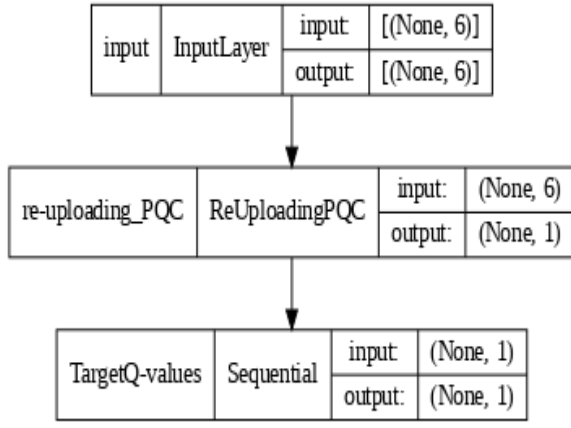
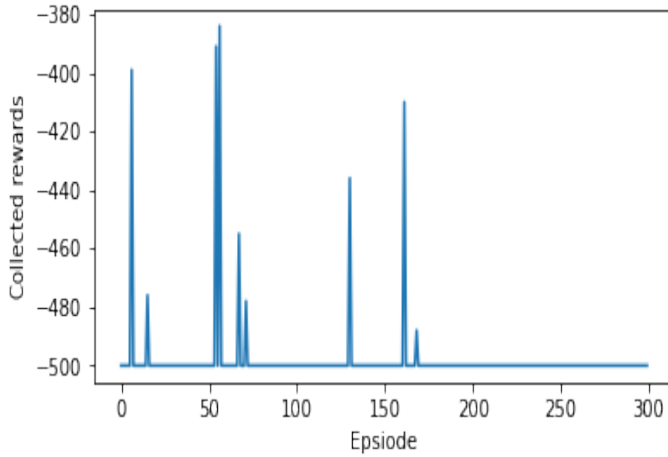
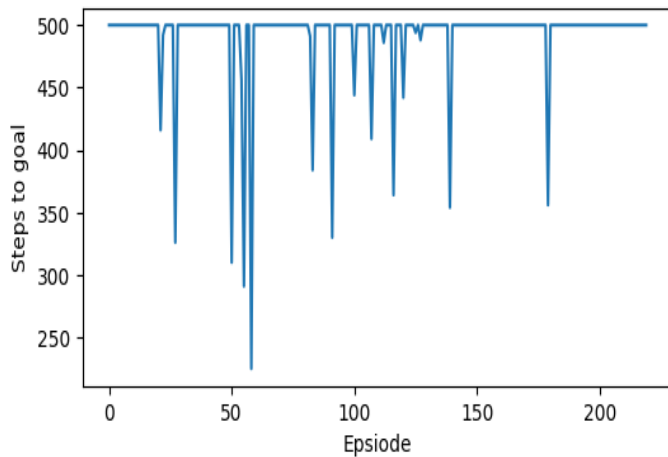


Fig. 7: PQC for Acrobot Q-function approximation.

the end goal within the episode, then the steps to reach the goal are plotted as 500. We observe that in the 200 episodes under consideration, the agent gets the end goal in roughly 20 episodes. The steps taken to reach the end goal also decrease as the number of episodes progresses.



(a) Average rewards by proposed DQL approach on Acrobot environment



(b) Steps to reach end goal for the agent in Acrobot environment

Fig. 8: Performance of Quantum Enhanced DQL on Acrobot environment

VI. CONCLUSION

This paper proposed a method to enhance the performance of Deep Q Learning. It was noticed that despite running on a simulated quantum environment in the absence of physical quantum hardware, the proposed model performed better than its classical counterpart. This implies that the proposed parameterized quantum circuits can be used in more Learning settings to enhance the performance of learning algorithms. Quantum Computing is an emergent field with many potential applications to overcome the limitations of present computing systems. This shows that future quantum computers can address various challenges in real-world machine-learning problems. The challenge is to pick the suitable quantum algorithm to help speed up the learning problem. Thus future work directs to search for algorithms and techniques in the quantum realm that can improve the solution's performance compared to the corresponding classical problem.

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